

Challenge Write-Up Template

Challenge Information

Student Name	Pascua, Diana KC S.
Section / Class	BSIT-3G
Challenge Name	MultiCode
Category	General Skills
Points / Difficulty	Easy
Date Solved	8/14/26
Flag Format Given	picoCTF{....}

1. Challenge Description

The challenge from picoCTF 2026 called "MultiCode,". The prompt was short and a little cryptic: "Can you peel back the layers and reveal the truth?" Along with it came a downloadable file. There was no server to connect to and no source code to read — just that one file. The task was basically to figure out what had been done to the message to scramble it, undo it, and recover a flag in the format picoCTF{...}.

2. Initial Analysis / Reconnaissance

The first thing I did was just open the message file and look at it. It was a single line of text made up of mixed upper- and lower-case letters and digits, and it ended in an equals sign. That '=' padding at the end is a dead giveaway for Base64, so that's where I decided to start. The challenge name itself "MultiCode", was also a hint that this probably wasn't going to be a single decode-and-done situation; it sounded like there would be more than one encoding stacked on top of each other. Given the file was just an obfuscated blob of text with no binary/executable behavior involved, I filed this under encoding/data-format rather than anything to do with crypto keys or networking.

3. Tools and Commands Used

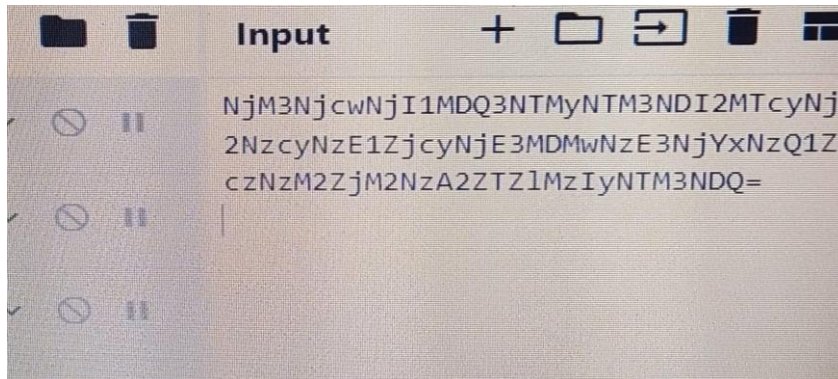
Tool / Command	What it was used for	Result / Output
Text Editor/Browser	Opened and inspected the downloaded message file.	A long block of mixed case letters, numbers, and trailing “=” signs. The classic look of base64.
CyberChef – from base64	First decoding pass on the raw message.	Output turned into a string of hex looking pairs(0-9, a-f)
CyberChef – from hex	Second decoding pass on the hex output	Output turned into text containing %-style escape sequences
CyberChef – from URLDecode	Third decoding pass to strip the percent encoding	Output turned into readable looking text, but still not the flag, letters were shifted
CyberChef – ROT13	Final decoding pass to undo the letter shift	Produced the plaintext flag.

4. Solution Walkthrough

Step 1

Identify and undo the base64 layer

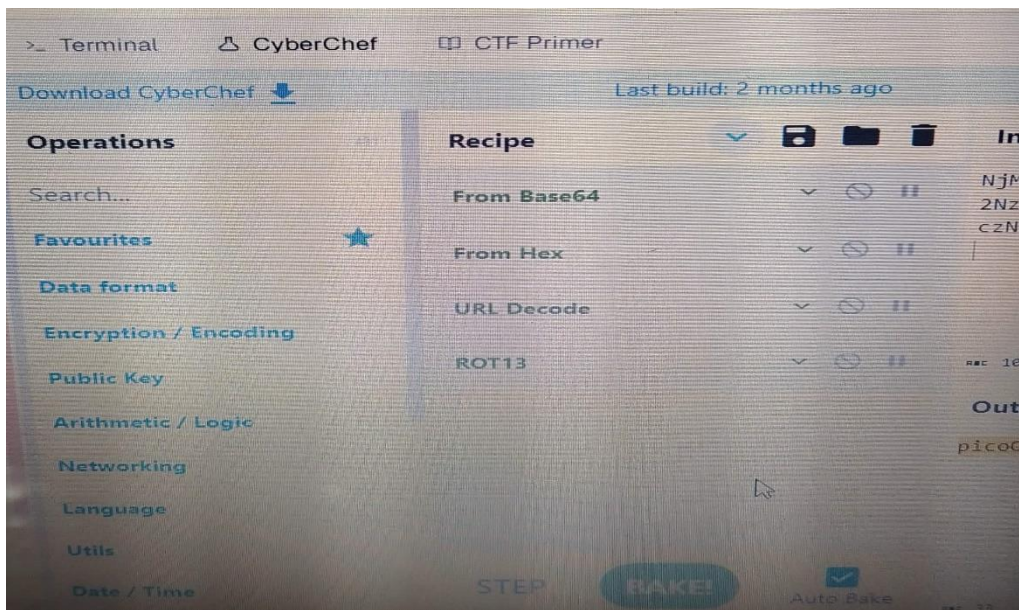
I pasted the contents of message into cyberchef's input pane and dragged in a "from Base64" operation. The output wasn't readable text, but it was clean- just pairs of characters from 0-9 and a-f, which is exactly what raw hex looks like. That told me there was another layer underneath.



Step 2

Identify and undo the hex layer

I added a "from hex" operation right after "from Base64". The output this time contained a bunch of percent signs followed by two characters each. That's the signature of URL/percent-encoding, so I knew what to add next.



Step 3

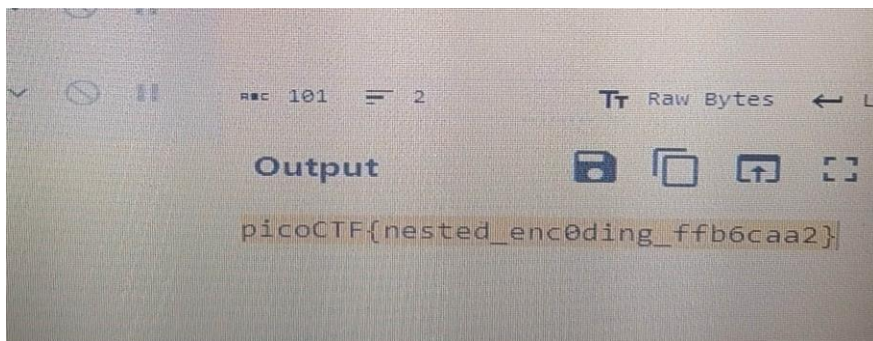
Undo the URL encoding

I added a “URL Decode” step. The output now looked like normal words and spacing but the letters were clearly shifted since the next wasn’t actually readable English and didn’t spell “picoCTF” at the start the way I expected the flag to.

Step 4

Recognize and undo the letter shift

Scrambled but still alphabetic text with the same word lengths and punctuation as normal English is the classic sign of a Caesar style shift cipher, and ROT13 specifically is the one you see constantly in these kinds of challenges because it’s its own inverse. I added a final “ROT13” operation to the recipe, and the output immediately resolved into clean, readable text starting with “picoCTF{”, confirming the whole chain was: Base64 – Hex – URL Decoding – ROT13, applied in that order to hide the flag, and I just had to peel them back in reverse.



5. Flag

picoCTF{nested_enc0ding_ffb6caa2}

6. Key Concept / What I Learned

This challenge was really recognizing encoding “signatures” by sight the “=” padding of Base64, the clean 0-9/a-f pattern of hex, the %XX pattern of URL encoding, and still alphabetic but wrong look of a Caesar/ROT shift and understanding that encodings can be stacked(nested) on top of each other. None of these individual steps are encryption in the security sense; they don’t need a secret key, so the real skills is pattern recognition and patience, decode one visible layer at a time, look at what the output resembles, and keep going until it stop changing shape and starts reading as plaintext.

7. References

CyberChef (Data-Format / Encryption-Encoding operation categories, including From Base64, From Hex, URL Decode, and ROT13)
